

MAEBASHI, Japan

WMO: 476240

Lat: 36.4053N

Lon: 139.0606E

Elev: 373

StdP: 14.50

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	27.3	29.1	8.5	8.6	41.2	11.0	9.8	39.3	20.3	44.0	18.4	43.2	6.6	320	0.396

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	14.6	95.3	77.0	92.7	76.3	90.0	75.3	78.8	90.2	77.8	88.7	76.8	87.0	7.5	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
75.8	136.9	83.9	74.7	131.8	83.1	73.7	127.4	82.4	42.6	90.1	41.5	89.1	40.5	87.2	82.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.3	14.8	13.0	DB	23.7	99.7	1.9	1.8	22.3	100.9	21.2	102.0	20.1	103.0	18.7	104.3
			WB	20.5	80.4	1.6	1.2	19.4	81.2	18.5	81.9	17.6	82.5	16.5	83.3

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	59.7	39.3	40.6	46.5	56.7	66.0	72.1	79.4	81.2	73.8	63.5	52.7	43.8	(6)
(7)		DBStd	15.44	4.03	4.79	5.82	6.37	5.34	5.08	5.63	5.00	5.77	5.30	5.24	4.80	(7)
(8)		HDD50	990	332	268	142	16	0	0	0	0	0	0	31	201	(8)
(9)		HDD65	3489	797	684	572	257	52	6	0	0	4	91	369	657	(9)
(10)		CDD50	4549	1	4	35	216	496	663	912	967	714	420	112	9	(10)
(11)		CDD65	1572	0	0	0	7	83	219	447	502	268	46	0	0	(11)
(12)		CDH74	13817	0	0	2	74	597	1470	4385	5242	1879	168	0	0	(12)
(13)		CDH80	5370	0	0	0	6	123	424	1913	2301	581	22	0	0	(13)
(14)	Wind	WSAvg	6.2	7.2	7.5	7.5	6.9	6.1	5.2	4.9	5.1	5.1	5.5	6.4	6.9	(14)
(15)	Precipitation	PrecAvg	46.5	1.0	1.2	2.1	3.0	3.8	6.1	7.2	7.4	7.6	4.4	1.8	0.9	(15)
(16)		PrecMax	65.3	4.1	6.2	4.8	6.9	10.0	12.9	14.8	22.0	17.7	15.5	7.8	4.5	(16)
(17)		PrecMin	31.3	0.0	0.0	0.0	0.2	0.8	1.8	1.8	0.9	0.8	0.2	0.0	0.0	(17)
(18)		PrecStd	7.9	1.0	1.2	1.1	1.5	1.9	2.6	2.9	3.8	3.9	3.1	1.4	0.9	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	57.9	63.1	71.4	80.2	87.4	92.5	98.2	98.4	92.7	83.1	71.2	62.8	(19)
(20)			MCWB	44.9	49.2	52.0	58.4	65.0	72.1	77.1	77.7	75.2	67.9	57.3	50.7	(20)
(21)		2%	DB	53.1	56.5	65.3	75.8	83.3	87.6	95.5	95.7	89.1	78.3	67.6	58.3	(21)
(22)			MCWB	41.7	44.0	49.7	57.3	63.3	69.6	77.1	77.4	74.0	65.3	55.6	47.2	(22)
(23)		5%	DB	49.9	52.8	61.1	72.1	80.2	84.7	92.8	93.3	86.2	75.1	64.6	55.2	(23)
(24)			MCWB	39.5	41.7	47.3	55.9	61.8	68.8	76.4	76.8	72.9	63.2	53.6	45.3	(24)
(25)		10%	DB	47.4	49.8	57.4	68.6	77.2	81.9	89.8	90.6	83.2	72.4	61.9	52.4	(25)
(26)			MCWB	38.0	39.8	46.2	54.5	61.1	67.8	75.6	76.2	71.4	61.7	52.2	43.3	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	46.7	50.7	55.3	65.5	70.3	75.7	80.1	80.6	77.5	70.9	61.1	53.1	(27)
(28)			MCD B	53.5	61.4	64.0	71.7	79.4	86.6	92.9	93.2	87.0	78.7	66.3	58.8	(28)
(29)		2%	WB	43.2	45.9	52.3	61.3	67.2	73.5	78.6	79.2	76.1	68.2	58.3	49.2	(29)
(30)			MCD B	51.6	53.7	61.7	71.7	77.3	82.8	90.6	90.5	84.7	75.5	63.7	55.8	(30)
(31)		5%	WB	40.5	43.1	49.6	58.7	65.1	71.9	77.6	78.2	74.9	65.9	56.2	46.5	(31)
(32)			MCD B	48.6	50.7	58.9	67.1	74.5	80.6	88.9	89.0	82.5	72.1	61.7	54.0	(32)
(33)		10%	WB	38.5	40.5	46.9	56.4	63.4	70.1	76.5	77.3	73.6	63.8	53.9	44.1	(33)
(34)			MCD B	46.2	48.7	56.0	64.9	72.8	78.0	87.0	87.7	80.8	70.4	60.1	51.3	(34)
(35)	Mean Daily Temperature Range		MDBR	15.7	16.4	17.4	18.5	17.5	14.6	14.3	14.6	13.4	14.2	15.7	15.8	(35)
(36)		5% DB	MCD BR	19.0	21.0	23.1	24.0	23.2	19.9	19.3	18.9	17.0	17.2	18.5	18.6	(36)
(37)			MCWBR	11.3	12.6	13.1	12.4	10.6	8.1	6.8	6.6	6.7	8.6	10.4	11.2	(37)
(38)			MCD BR	17.8	19.1	19.8	18.5	17.9	15.3	17.2	16.8	14.3	13.8	14.7	17.0	(38)
(39)		5% WB	MCWBR	11.4	12.6	12.7	11.6	10.0	7.0	6.5	6.4	6.3	7.9	9.7	11.1	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.373	0.420	0.494	0.550	0.583	0.627	0.620	0.621	0.534	0.478	0.410	0.373	(40)	
(41)		taud	2.110	1.976	1.869	1.823	1.799	1.764	1.834	1.823	2.002	2.075	2.190	2.198	(41)	
(42)		Ebn at Noon	250	251	243	238	232	221	222	218	230	231	236	242	(42)	
(43)		Edh at Noon	38	49	60	66	69	71	66	65	52	44	35	33	(43)	
(44)	All-Sky Solar Radiation	RadAvg	792	1020	1294	1520	1654	1486	1477	1474	1189	986	797	705	(44)	
(45)		RadStd	43	68	77	126	166	111	221	151	101	96	64	49	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (43 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page